A6

Q1

$$\nabla f(x) = \begin{bmatrix} 3x_1^2 - 36x_1 - 84 \\ 3x_2^2 - 36x_2 \end{bmatrix}$$

Set the gradient equal to zero:

1. $3x_1^2 - 36x_1 - 84 = 0$
2. $3x_2^2 - 36x_2 = 0$

$\Rightarrow$ The critical points are all combinations of x_1 and x_2:

1. $(14, 0)$
2. $(14, 12)$
3. $(-2, 0)$
4. $(-2, 12)$

The Hessian matrix is:

$$H(x) = \begin{bmatrix} 6x_1 - 36 & 0 \\ 0 & 6x_2 - 36 \end{bmatrix}$$

Evaluate H at each critical point:

1. $(14, 0)$:

$$H(14, 0) = \begin{bmatrix} 6 \cdot 14 - 36 & 0 \\ 0 & 6 \cdot 0 - 36 \end{bmatrix} = \begin{bmatrix} 48 & 0 \\ 0 & -36 \end{bmatrix}$$

   Eigenvalues: 48 (> 0), -36 (< 0). The Hessian is indefinite.
   - **Conclusion**: Saddle point.

2. $(14, 12)$:

$$H(14, 12) = \begin{bmatrix} 6 \cdot 14 - 36 & 0 \\ 0 & 6 \cdot 12 - 36 \end{bmatrix} = \begin{bmatrix} 48 & 0 \\ 0 & 36 \end{bmatrix}$$

   Eigenvalues: 48 (> 0), 36 (> 0). The Hessian is positive definite.
   - **SONC**: Positive semi-definite (satisfied).
   - **SOSC**: Positive definite implies a strict local minimizer.
   - **Conclusion**: Strict local minimizer.

3. $(-2, 0)$:

$$H(-2, 0) = \begin{bmatrix} 6 \cdot (-2) - 36 & 0 \\ 0 & 6 \cdot 0 - 36 \end{bmatrix} = \begin{bmatrix} -48 & 0 \\ 0 & -36 \end{bmatrix}$$

   Eigenvalues: -48 (< 0), -36 (< 0). The Hessian is negative definite.
   - **SONC**: Negative semi-definite (satisfied).
   - **SOSC**: Negative definite implies a strict local maximizer.
   - **Conclusion**: Strict local maximizer.

4. $(-2, 12)$:

$$H(-2, 12) = \begin{bmatrix} 6 \cdot (-2) - 36 & 0 \\ 0 & 6 \cdot 12 - 36 \end{bmatrix} = \begin{bmatrix} -48 & 0 \\ 0 & 36 \end{bmatrix}$$

   Eigenvalues: -48 (< 0), 36 (> 0). The Hessian is indefinite.
   - **Conclusion**: Saddle point.

Final Answer

a) **Saddle points**: $(14, 0)$ and $(-2, 12)$
b) **Strict local minimizers**: $(14, 12)$
c) **Strict local maximizers**: $(-2, 0)$

Q2

Part (a): KKT Conditions

The Lagrangian is:

$$\mathcal{L}(x, \lambda, \mu) = x^T Ax + \lambda(x^T x - 2) + \mu(x_1 - 1)$$

where λ and μ are Lagrange multipliers for the two constraints.

The KKT conditions require:

1. **Stationarity**: The gradient of the Lagrangian with respect to x is zero:

$$\nabla_x \mathcal{L} = 0$$

2. **Primal Feasibility**: The constraints are satisfied:

$$x^T x = 2, \quad x_1 = 1$$

Compute the gradient:

- $\nabla_x (x^T Ax) = 2Ax$ (since A is symmetric)
- $\nabla_x (x^T x - 2) = 2x$
- $\nabla_x (x_1 - 1) = e_1$ (where $e_1 = (1, 0, \ldots, 0)^T$ is the first standard basis vector, because $x_1 = e_1^T x$)

So the KKT conditions are:

$$2Ax + 2\lambda x + \mu e_1 = 0$$
$$x^T x = 2$$
$$x_1 = 1$$

Part (b): KKT Points When A is Diagonal

Start with the KKT conditions:

- Stationarity: $2Ax + 2\lambda x + \mu e_1 = 0$
- Constraints: $x^T x = 2$, $x_1 = 1$

$\because$ A is diagonal
$\therefore$ For the i -th component:

$$2a_i x_i + 2\lambda x_i + \mu \delta_{i1} = 0$$

where $\delta_{i1} = 1$ if $i = 1$, and 0 otherwise.

- For $i = 1$:

$$2a_1 x_1 + 2\lambda x_1 + \mu = 0$$

  With $x_1 = 1$:

$$2a_1 + 2\lambda + \mu = 0 \implies \mu = -2a_1 - 2\lambda$$

- For $i \geq 2$:

$$2a_i x_i + 2\lambda x_i = 0 \implies 2x_i(a_i + \lambda) = 0$$

  Thus, either:
  - $x_i = 0$, or
  - $a_i + \lambda = 0 \implies \lambda = -a_i$

Since λ is a single scalar and the a_i 's are distinct, $\lambda = -a_i$ can hold for at most one i, because if $\lambda = -a_i = -a_j$ for $i \neq j$, then $a_i = a_j$, contradicting distinctness.

Continue using the constraint $x^T x = 2$:

With $x_1 = 1$:

$$x_1^2 + x_2^2 + \cdots + x_n^2 = 1 + \sum_{i=2}^{n} x_i^2 = 2 \implies \sum_{i=2}^{n} x_i^2 = 1$$

So, $(x_2, \ldots, x_n)$ lies on the unit sphere in $\mathbb{R}^{n-1}$. Now, consider the stationarity for $i \geq 2$:

- If $x_i \neq 0$, then $\lambda = -a_i$, and for all $j \geq 2$, $j \neq i$, $2x_j(a_j + \lambda) = 2x_j(a_j - a_i) = 0$. Since $a_j \neq a_i$ (distinctness), $x_j = 0$.

Thus, at most one x_i for $i \geq 2$ can be nonzero. Test this:

- **Case: All $x_i = 0$ for $i \geq 2$**
  Then $x = (1, 0, \ldots, 0)^T$, and $x^T x = 1 \neq 2$, which violates the constraint. This case is infeasible for $n \geq 2$. (If $n = 1$, $x_1 = 1$ gives $x^T x = 1 \neq 2$, so no solution exists; assume $n \geq 2$.)
- **Case: Exactly one $x_k \neq 0$ for some $k \geq 2$**
  Let $x_k \neq 0$, and $x_i = 0$ for $i \geq 2$, $i \neq k$. Then:

$$x_1^2 + x_k^2 = 1 + x_k^2 = 2 \implies x_k^2 = 1 \implies x_k = \pm 1$$

  Stationarity for $i = k$: $2a_k x_k + 2\lambda x_k = 0 \implies a_k + \lambda = 0 \implies \lambda = -a_k$.
  For $i \geq 2$, $i \neq k$: $x_i = 0$, so $2x_i(a_i + \lambda) = 0$ holds.
  For $i = 1$: $\mu = -2a_1 - 2\lambda = -2a_1 + 2a_k = 2(a_k - a_1)$.

Verify:

- $x = (1, 0, \ldots, x_k = \pm 1, \ldots, 0)^T$, $\lambda = -a_k$, $\mu = 2a_k - 2a_1$
- $x^T x = 2$
- $x_1 = 1$
- Stationarity holds component-wise.

For each $k = 2, \ldots, n$, there are two points ($x_k = 1$ or -1). Total KKT points: $2(n - 1)$.

KKT Points: For each $k = 2, \ldots, n$ and $s = \pm 1$, $x_1 = 1$, $x_k = s$, $x_i = 0$ for $i \neq 1, k$, with $\lambda = -a_k$, $\mu = 2a_k - 2a_1$.
Number: $2(n - 1)$.

Part (c): Optimal Value

For $x = (1, 0, \ldots, x_k = \pm 1, \ldots, 0)^T$:

$$x^T A x = a_1 x_1^2 + a_k x_k^2 = a_1 + a_k$$

The value depends on k, not s, and ranges over $a_1 + a_k$ for $k = 2, \ldots, n$. Since $a_i > 0$ and distinct, the minimum is:

$$a_1 + \min_{k=2,\ldots,n} a_k$$

Let $a_m = \min_{i=2,\ldots,n} a_i$. The optimal value is $a_1 + a_m$, achieved at $k = m$ with $x_m = \pm 1$.

Since A is positive definite, $x^T A x$ is strictly convex, and the feasible set is compact (a sphere intersected with a hyperplane), the minimum exists and is among the KKT points. This confirms $a_1 + a_m$ as the optimal value.

Final Answer

- **(a)** The KKT conditions are:

$$2Ax + 2\lambda x + \mu e_1 = 0, \quad x^T x = 2, \quad x_1 = 1$$

- **(b)** For $A = \text{diag}(a_1, \ldots, a_n)$ with distinct $a_i > 0$, there are $2(n - 1)$ KKT points. For each $k = 2, \ldots, n$, the point has $x_1 = 1$, $x_k = \pm 1$, $x_i = 0$ for $i \neq 1, k$, with $\lambda = -a_k$, $\mu = 2a_k - 2a_1$.
- **(c)** The optimal value is:

$$a_1 + \min_{i=2,\ldots,n} a_i$$

Q3

Standardize the Constraints

The problem becomes:
Maximize $f(\mathbf{x}) = 4x_1 + 3x_2 + 6x_3$
Subject to:

- $g_1(\mathbf{x}) = 5x_1 + 2x_2 + 3x_3 - 15 \leq 0$
- $g_2(\mathbf{x}) = 3x_1 + 4x_2 + x_3 - 9 \leq 0$
- $g_3(\mathbf{x}) = -x_1 - x_2 - x_3 \leq 0$

Define the KKT Conditions

The stationarity condition becomes:

$$-\nabla f(\mathbf{x}) + \sum_i \mu_i \nabla g_i(\mathbf{x}) = 0$$

Or equivalently:

$$\nabla f(\mathbf{x}) = \sum_i \mu_i \nabla g_i(\mathbf{x})$$

where $\mu_i \geq 0$ are the Lagrange multipliers associated with each constraint $g_i(\mathbf{x}) \leq 0$. The full KKT conditions are:

1. **Stationarity**: $\nabla f(\mathbf{x}) = \mu_1 \nabla g_1(\mathbf{x}) + \mu_2 \nabla g_2(\mathbf{x}) + \mu_3 \nabla g_3(\mathbf{x})$
2. **Primal Feasibility**: $g_i(\mathbf{x}) \leq 0$ for $i = 1, 2, 3$
3. **Dual Feasibility**: $\mu_i \geq 0$ for $i = 1, 2, 3$
4. **Complementary Slackness**: $\mu_i g_i(\mathbf{x}) = 0$ for $i = 1, 2, 3$

Compute Gradients

- Objective function: $f(\mathbf{x}) = 4x_1 + 3x_2 + 6x_3$

$$\nabla f = (4, 3, 6)$$

- Constraint 1: $g_1(\mathbf{x}) = 5x_1 + 2x_2 + 3x_3 - 15$

$$\nabla g_1 = (5, 2, 3)$$

- Constraint 2: $g_2(\mathbf{x}) = 3x_1 + 4x_2 + x_3 - 9$

$$\nabla g_2 = (3, 4, 1)$$

- Constraint 3: $g_3(\mathbf{x}) = -x_1 - x_2 - x_3$

$$\nabla g_3 = (-1, -1, -1)$$

The KKT Conditions

Stationarity

$$(4, 3, 6) = \mu_1(5, 2, 3) + \mu_2(3, 4, 1) + \mu_3(-1, -1, -1)$$

Breaking this into components:

- x_1-direction: $5\mu_1 + 3\mu_2 - \mu_3 = 4$
- x_2-direction: $2\mu_1 + 4\mu_2 - \mu_3 = 3$
- x_3-direction: $3\mu_1 + \mu_2 - \mu_3 = 6$

Primal Feasibility

- $g_1(\mathbf{x}) \leq 0$: $5x_1 + 2x_2 + 3x_3 - 15 \leq 0$ (or $5x_1 + 2x_2 + 3x_3 \leq 15$)
- $g_2(\mathbf{x}) \leq 0$: $3x_1 + 4x_2 + x_3 - 9 \leq 0$ (or $3x_1 + 4x_2 + x_3 \leq 9$)
- $g_3(\mathbf{x}) \leq 0$: $-x_1 - x_2 - x_3 \leq 0$ (or $x_1 + x_2 + x_3 \geq 0$)

Dual Feasibility

- $\mu_1 \geq 0$
- $\mu_2 \geq 0$
- $\mu_3 \geq 0$

Complementary Slackness

- $\mu_1 g_1(\mathbf{x}) = 0$: $\mu_1(5x_1 + 2x_2 + 3x_3 - 15) = 0$
- $\mu_2 g_2(\mathbf{x}) = 0$: $\mu_2(3x_1 + 4x_2 + x_3 - 9) = 0$
- $\mu_3 g_3(\mathbf{x}) = 0$: $\mu_3(-x_1 - x_2 - x_3) = 0$

Final KKT Conditions

The complete set of KKT conditions for the given linear program is:

- **Stationarity**:
  - $5\mu_1 + 3\mu_2 - \mu_3 = 4$
  - $2\mu_1 + 4\mu_2 - \mu_3 = 3$
  - $3\mu_1 + \mu_2 - \mu_3 = 6$
- **Primal Feasibility**:
  - $5x_1 + 2x_2 + 3x_3 \leq 15$
  - $3x_1 + 4x_2 + x_3 \leq 9$
  - $-x_1 - x_2 - x_3 \leq 0$ (equivalently, $x_1 + x_2 + x_3 \geq 0$)
- **Dual Feasibility**:

- $\mu_1 \geq 0$
- $\mu_2 \geq 0$
- $\mu_3 \geq 0$

- **Complementary Slackness**:
  - $\mu_1(5x_1 + 2x_2 + 3x_3 - 15) = 0$
  - $\mu_2(3x_1 + 4x_2 + x_3 - 9) = 0$
  - $\mu_3(-x_1 - x_2 - x_3) = 0$

Q4

Standardize the Constraints

Convert each constraint:

1. **Constraint** $x_1 + x_2 \leq 4$:
   - $g_1(x) = x_1 + x_2 - 4 \leq 0$
2. **Constraint** $x_3 - x_2^2 \geq 2$:
   - $g_2(x) = 2 + x_2^2 - x_3 \leq 0$
3. **Non-negativity constraints**:
   - $x_1 \geq 0$: $g_3(x) = -x_1 \leq 0$
   - $x_2 \geq 0$: $g_4(x) = -x_2 \leq 0$
   - $x_3 \geq 0$: $g_5(x) = -x_3 \leq 0$

Define the Lagrangian

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^{5} \lambda_i g_i(x)$$

Substitute the expressions:

$$\mathcal{L}(x, \lambda) = x_1 e^{x_1} + (x_2 - 1)^2 + x_3^2 + \lambda_1(x_1 + x_2 - 4) + \lambda_2(2 + x_2^2 - x_3) + \lambda_3(-x_1) + \lambda_4(-x_2) + \lambda_5(-x_3)$$

KKT Conditions

1. **Stationarity**: $\nabla_x \mathcal{L}(x, \lambda) = 0$
2. **Primal feasibility**: $g_i(x) \leq 0$ for $i = 1, 2, 3, 4, 5$
3. **Dual feasibility**: $\lambda_i \geq 0$ for $i = 1, 2, 3, 4, 5$
4. **Complementary slackness**: $\lambda_i g_i(x) = 0$ for $i = 1, 2, 3, 4, 5$

Stationarity

Compute the partial derivatives of $\mathcal{L}$ with respect to x_1, x_2, x_3 and set them to zero.

- **With respect to** x_1:

$$\frac{\partial \mathcal{L}}{\partial x_1} = e^{x_1}(1 + x_1) + \lambda_1 - \lambda_3 = 0$$

- **With respect to** x_2:

$$\frac{\partial \mathcal{L}}{\partial x_2} = 2(x_2 - 1) + \lambda_1 + \lambda_2 \cdot 2x_2 - \lambda_4 = 0$$

- **With respect to** x_3:

$$\frac{\partial \mathcal{L}}{\partial x_3} = 2x_3 + \lambda_2 \cdot (-1) - \lambda_5 = 2x_3 - \lambda_2 - \lambda_5 = 0$$

So, the stationarity conditions are:

$$e^{x_1}(1 + x_1) + \lambda_1 - \lambda_3 = 0$$

$$2(x_2 - 1) + \lambda_1 + 2\lambda_2 x_2 - \lambda_4 = 0$$

$$2x_3 - \lambda_2 - \lambda_5 = 0$$

Primal Feasibility

Using the original constraints for clarity:

$$x_1 + x_2 \leq 4$$

$$x_3 - x_2^2 \geq 2$$

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0$$

Dual Feasibility

$$\lambda_1 \geq 0, \quad \lambda_2 \geq 0, \quad \lambda_3 \geq 0, \quad \lambda_4 \geq 0, \quad \lambda_5 \geq 0$$

Complementary Slackness

$$\lambda_1 g_1(x) = \lambda_1(x_1 + x_2 - 4) = 0$$

$$\lambda_2 g_2(x) = \lambda_2(2 + x_2^2 - x_3) = 0$$

$$\lambda_3 g_3(x) = \lambda_3(-x_1) = 0$$

$$\lambda_4 g_4(x) = \lambda_4(-x_2) = 0$$

$$\lambda_5 g_5(x) = \lambda_5(-x_3) = 0$$

Since $x_1, x_2, x_3 \geq 0$ and $\lambda_3, \lambda_4, \lambda_5 \geq 0$, these can also be written as:

$$\lambda_3 x_1 = 0$$

$$\lambda_4 x_2 = 0$$

$$\lambda_5 x_3 = 0$$

For the second constraint, note that $2 + x_2^2 - x_3 = 0$ implies $x_3 = x_2^2 + 2$, which satisfies $x_3 - x_2^2 - 2 = 0$, so alternatively, we could write $\lambda_2(x_3 - x_2^2 - 2) = 0$, aligning with the original constraint form.

Final KKT Conditions

Combining all conditions, and presenting them clearly with the original constraint forms where appropriate:

- **Stationarity**:
  - $e^{x_1}(1 + x_1) + \lambda_1 - \lambda_3 = 0$
  - $2(x_2 - 1) + \lambda_1 + 2\lambda_2 x_2 - \lambda_4 = 0$
  - $2x_3 - \lambda_2 - \lambda_5 = 0$
- **Primal feasibility**:
  - $x_1 + x_2 \leq 4$
  - $x_3 \geq x_2^2 + 2$
  - $x_1 \geq 0$
  - $x_2 \geq 0$
  - $x_3 \geq 0$
- **Dual feasibility**:
  - $\lambda_1 \geq 0$
  - $\lambda_2 \geq 0$
  - $\lambda_3 \geq 0$
  - $\lambda_4 \geq 0$
  - $\lambda_5 \geq 0$
- **Complementary slackness**:
  - $\lambda_1(x_1 + x_2 - 4) = 0$
  - $\lambda_2(x_3 - x_2^2 - 2) = 0$ (since $x_3 - x_2^2 \geq 2$, equality holds when the constraint is active)
  - $\lambda_3 x_1 = 0$
  - $\lambda_4 x_2 = 0$
  - $\lambda_5 x_3 = 0$

Q5

Part 1: SOSC Implies QGC

Assume that SOSC holds at x^*:

- $\nabla f(x^*) = 0$,
- $\nabla^2 f(x^*)$ is positive definite.

Now need to show that there exist $\alpha > 0$ and $\epsilon > 0$ such that for all $x \in B_\epsilon(x^*)$,

$$f(x) - f(x^*) \geq \frac{\alpha}{2}\|x - x^*\|^2.$$

Since f is twice continuously differentiable, use the second-order Taylor expansion around x^*. For any x near x^*, there exists a point ξ on the line segment between x^* and x such that:

$$f(x) = f(x^*) + \nabla f(x^*)^T(x - x^*) + \frac{1}{2}(x - x^*)^T\nabla^2 f(\xi)(x - x^*).$$

Given $\nabla f(x^*) = 0$, this simplifies to:

$$f(x) - f(x^*) = \frac{1}{2}(x - x^*)^T\nabla^2 f(\xi)(x - x^*).$$

Now, since $\nabla^2 f(x^*)$ is positive definite, its smallest eigenvalue, $\lambda_{\min}(\nabla^2 f(x^*))$, is positive: $\lambda_{\min}(\nabla^2 f(x^*)) > 0$. Because f is C^2, the Hessian $\nabla^2 f(x)$ is continuous as a function of x, and the eigenvalues of a symmetric matrix are continuous functions of its entries. Thus, $\lambda_{\min}(\nabla^2 f(x))$ is continuous in x.

Since $\lambda_{\min}(\nabla^2 f(x^*)) > 0$, by continuity, there exists an $\epsilon > 0$ such that for all $x \in B_\epsilon(x^*)$,

$$\lambda_{\min}(\nabla^2 f(x)) \geq \frac{1}{2}\lambda_{\min}(\nabla^2 f(x^*)) > 0.$$

This ensures that $\nabla^2 f(x)$ remains positive definite in $B_\epsilon(x^*)$, with its smallest eigenvalue bounded below by a positive constant. Define:

$$\alpha = \frac{1}{2}\lambda_{\min}(\nabla^2 f(x^*)).$$

For any $x \in B_\epsilon(x^*)$, since ξ lies on the segment from x^* to x, we have $\xi \in B_\epsilon(x^*)$ (because $B_\epsilon(x^*)$ is convex). Thus:

$$\lambda_{\min}(\nabla^2 f(\xi)) \geq \alpha > 0.$$

For a symmetric positive definite matrix A, the quadratic form satisfies:

$$v^T A v \geq \lambda_{\min}(A)\|v\|^2.$$

Applying this to $\nabla^2 f(\xi)$ with $v = x - x^*$,

$$(x - x^*)^T\nabla^2 f(\xi)(x - x^*) \geq \lambda_{\min}(\nabla^2 f(\xi))\|x - x^*\|^2 \geq \alpha\|x - x^*\|^2.$$

Substituting into the Taylor expansion:

$$f(x) - f(x^*) = \frac{1}{2}(x - x^*)^T\nabla^2 f(\xi)(x - x^*) \geq \frac{1}{2}\alpha\|x - x^*\|^2.$$

This matches the QGC form exactly, with $\alpha > 0$ and $\epsilon > 0$ as required. Thus, SOSC implies QGC.

Part 2: QGC Implies SOSC

Assume that QGC holds: there exist $\alpha > 0$ and $\epsilon > 0$ such that for all $x \in B_\epsilon(x^*)$,

$$f(x) - f(x^*) \geq \frac{\alpha}{2}\|x - x^*\|^2.$$

Now need to prove:

1. $\nabla f(x^*) = 0$,
2. $\nabla^2 f(x^*)$ is positive definite.

Step 1: Show $\nabla f(x^*) = 0$

Since $\frac{\alpha}{2}\|x - x^*\|^2 \geq 0$ and is zero only when $x = x^*$, the inequality implies:

$$f(x) \geq f(x^*) \text{ for all } x \in B_\epsilon(x^*),$$

with equality at $x = x^*$. Thus, x^* is a local minimizer of f. For a C^2 function at a local minimizer, the gradient must be zero. To confirm, suppose $\nabla f(x^*) \neq 0$. Consider the direction $v = -\nabla f(x^*)$, and for small $t > 0$, set $x = x^* + tv$. By the first-order Taylor expansion:

$$f(x^* + tv) = f(x^*) + t\nabla f(x^*)^T v + o(t).$$

Since $v = -\nabla f(x^*)$,

$$\nabla f(x^*)^T v = -\nabla f(x^*)^T\nabla f(x^*) = -\|\nabla f(x^*)\|^2 < 0.$$

Thus, for small $t > 0$,

$$f(x^* + tv) < f(x^*),$$

contradicting the fact that $f(x) \geq f(x^*)$ in $B_\epsilon(x^*)$. Hence, $\nabla f(x^*) = 0$.

Step 2: Show $\nabla^2 f(x^*)$ is Positive Definite

With $\nabla f(x^*) = 0$, consider the second-order Taylor expansion for any $v \in \mathbb{R}^n$ and small $t > 0$:

$$f(x^* + tv) = f(x^*) + t\nabla f(x^*)^T v + \frac{1}{2}t^2 v^T \nabla^2 f(x^*) v + o(t^2).$$

Since $\nabla f(x^*) = 0$,

$$f(x^* + tv) - f(x^*) = \frac{1}{2}t^2 v^T \nabla^2 f(x^*) v + o(t^2).$$

By QGC, for t small enough that $x^* + tv \in B_\epsilon(x^*)$,

$$f(x^* + tv) - f(x^*) \geq \frac{\alpha}{2}\|x^* + tv - x^*\|^2 = \frac{\alpha}{2}t^2\|v\|^2.$$

Thus:

$$\frac{1}{2}t^2 v^T \nabla^2 f(x^*) v + o(t^2) \geq \frac{\alpha}{2}t^2\|v\|^2.$$

Divide by $t^2 > 0$:

$$\frac{1}{2}v^T \nabla^2 f(x^*) v + \frac{o(t^2)}{t^2} \geq \frac{\alpha}{2}\|v\|^2.$$

As $t \to 0$, $\frac{o(t^2)}{t^2} \to 0$, so:

$$\frac{1}{2}v^T \nabla^2 f(x^*) v \geq \frac{\alpha}{2}\|v\|^2.$$

Multiply through by 2:

$$v^T \nabla^2 f(x^*) v \geq \alpha\|v\|^2.$$

Since $\alpha > 0$, for all $v \neq 0$,

$$v^T \nabla^2 f(x^*) v \geq \alpha\|v^2 > 0,$$

which is the definition of positive definiteness. Alternatively, this implies that the smallest eigenvalue of $\nabla^2 f(x^*)$ is at least $\alpha > 0$, confirming that $\nabla^2 f(x^*)$ is positive definite.

Thus, QGC implies SOSC.